

Research Method of Data Science

Week 2: From a Broad Problem to a Researchable Question

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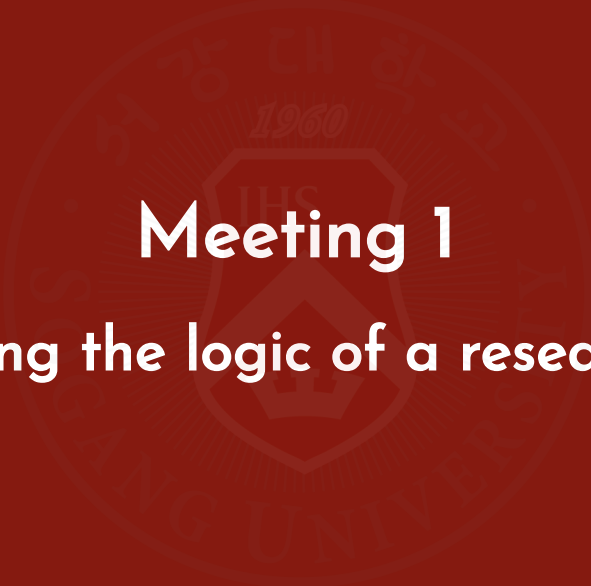
September 8/10, 2026

Welcome to Week 2!

From a Broad Topic to a Researchable Question

This week's output: A preliminary question that is clear, meaningful, and feasible enough to investigate.



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Meeting 1

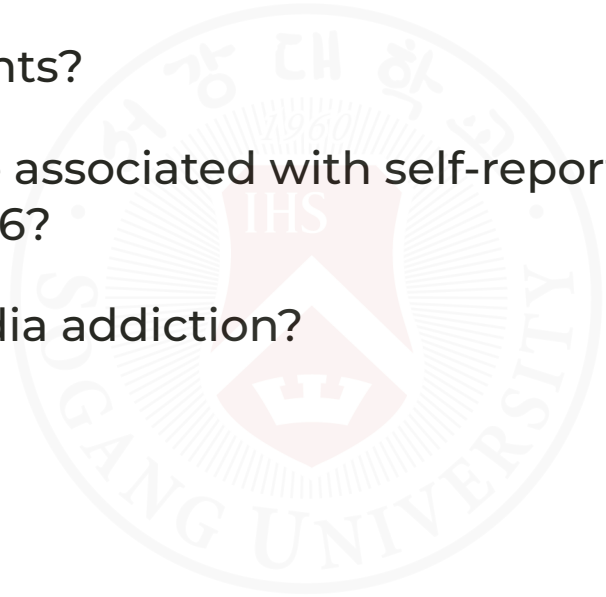
Understanding the logic of a research question

Opening diagnostic

Which is the strongest **research question**?

- A. Social media and mental health
- B. Is social media bad for students?
- C. How is daily social-media use associated with self-reported anxiety among Sogang undergraduates during Fall 2026?
- D. How can we solve social-media addiction?

Vote first. Then explain **why**.



Agenda

Week 2 — Meeting 1

1. From Week 1 to Week 2
 2. The Research Process
 3. From Subject and Topic to a Research Problem
 4. Developing a Researchable Question
 5. From Question to Evidence
 6. Worked Research Example
 7. Meeting 1 Summary
- Appendix



Learning objectives

By the end of this meeting, you should be able to:

1. explain why research is iterative;
2. connect a topic, problem, question, objective, and contribution;
3. formulate a bounded research question;
4. distinguish a concept from its measurement;
5. evaluate a question using **FINER**;
6. identify evidence aligned with the intended claim.

Our course framework

Question → Literature → Design → Evidence → Analysis → Claim

In Week 1, we learned that:

- the question constrains the evidence and method;
- research **approach** and research **purpose** are different dimensions;
- methods are tools—not rankings;
- the claim must not be stronger than the design permits.

This week asks: Where does a good research question come from?

The Week 1 AI-and-grades example is revisited in Appendix A.

Research is systematic—but iterative

Problem ↔ Literature ↔ Question ↔ Design ↔ Evidence ↔ Analysis ↔ Claim

Researchers often move backward:

- literature shows that the question is already answered;
- available evidence requires a narrower question;
- measurement problems force us to redefine a concept;
- preliminary analysis exposes a design weakness;
- results require a more modest interpretation.

Revision is part of rigorous research—not evidence that the research failed.

A step-by-step research cycle and an alternative visual model are provided in Appendix A.

Quick interaction: where would you go back?

What should the researcher reconsider?

1. The literature reveals an almost identical published study.
2. The desired survey population cannot be accessed.
3. The available variable measures income, not wealth.
4. The analysis shows association, but the draft says “causes.”

For each case, name the earlier research decision that must be revisited.

One connected chain

Topic → Problem / Puzzle → Research Question → Objective → Contribution

Each element answers a different question:

Element	Central question
Topic	What broad area interests us?
Problem / puzzle	What is not adequately understood?
Research question	What exactly will this study answer?
Objective	What will the study do to answer it?
Contribution	What new value could the answer add?

From subject to topic to question

- **Subject:** a broad area of knowledge—for example, central banking or climate change.
- **Topic:** a specific interest within that subject on which you could become a local expert—for example, the effect of central-bank communication on financial markets.
 - Narrow the topic until it is productively focused.
 - Try using an **action word**: affecting, changing, conflicting, developing, comparing.
 - Treat it as a path to pursue, not yet as the final research project.
- **Question:** what you want to find out about the topic whose answer addresses a problem that readers care about.
 - It directs you toward the information you need.
 - **How** and **why** questions often invite deeper thinking.
 - It should help readers see the topic in a new way.

See the Appendix for the topic → question → significance diagram.

Significance of problems in research

A research project needs a **problem** that its answer helps address.

- Without a problem, research easily becomes a data dump.
- Defining the problem gives the study focus and significance.
- Research problems may begin as **practical** or **conceptual** problems.
- Both have a two-part structure:
 1. a **condition**;
 2. an undesirable **consequence or cost** created by that condition.

The problem explains both **why the question is needed** and **why readers should care about the answer**.

Conceptual and practical problems

Conceptual problem

We do not understand something, and that gap prevents a more important understanding.

Research responds by **improving understanding**.

Academic research usually addresses a conceptual problem directly, even when motivated by a practical problem.

Practical problem

A condition in the world imposes tangible costs such as time, money, opportunity, security, health, or social harm.

Action responds by **changing the condition**.

Visual templates and fuller definitions are provided in Appendix A.

How the two problems connect

Practical problem:

- | Bank failures impose losses and threaten financial stability.

Conceptual problem:

- | We do not adequately understand which lending practices generate persistent loan-quality problems.

Research question:

- | How is related lending connected to nonperforming loans in Ukrainian banks?

Academic research usually addresses the **conceptual problem directly**. Better understanding may then help decision-makers address the practical problem.

How to find your research problem

- Look for a small problem inside a larger one.
- Read critically: find disagreements, unresolved puzzles, or findings that conflict with real-world knowledge.
- Question assumptions: **What if this assumption is wrong?**
- Ask whether an established result applies to another population, institution, place, or period.
- Keep asking **So what?**

It is acceptable not to have the final problem at the beginning. Continue testing and refining it throughout the research process.

See the Appendix for an additional visual summary of why problem definition matters.

Where do literature and theory enter?

They connect the topic to the problem.

Literature helps us establish:

- what is already known;
- where findings disagree;
- which contexts, groups, mechanisms, or methods remain underexamined.

Theory helps us explain:

- why concepts may be connected;
- which mechanism and comparison matter;
- what we should expect to observe.

Element 3: Research question—state the answer searched for

A research question converts the problem into a bounded inquiry.

It normally identifies:

- the phenomenon or relationship;
- the unit, population, or cases;
- the context or place;
- the period, when relevant;
- the intended type of claim.

| The wording of the question should match the claim the design can support.

Concepts inside the question

Consider:

| Does institutional quality increase economic development?

What do we mean by:

- **institutional quality?**
- **economic development?**
- **increase?**
- national or subnational institutions?
- short-run or long-run change?

Conceptualization: What exactly do we mean?

Operationalization: How will we observe or measure it?



Question and objective must align

If the question asks...	The objective might be...
What is happening?	to describe the pattern or distribution
How is X related to Y?	to estimate and interpret an association
Why or how does it occur?	to examine mechanisms or explanations
What is the effect of X?	to estimate a causal effect
What will happen?	to develop and evaluate a prediction

An objective should not promise more than the question or design permits.

See Appendix A for a complete question-to-objective example.

Follow one example through the chain

Element	AI-learning example
Topic	generative AI in university education
Problem	higher AI-assisted task scores may not represent retained learning
Question	Does access to an AI tutor affect students' later unaided test performance?
Objective	estimate the effect of AI-tutor access on later unaided performance
Contribution	distinguish immediate task assistance from retained learning

The five elements are not five unrelated labels. Each one narrows and justifies the next.

Two-minute chain test

With a partner, complete the missing elements:

Element	Draft
Topic	remote work
Problem	?
Question	?
Objective	?
Contribution	?

There is no single correct answer—but the five elements must form one coherent study.

Anatomy of a researchable question

For a relationship:

| How / why / to what extent does **[X]** relate to, explain, affect, or predict **[Y]** among **[population]** in **[context/period]**?

For an experience or process:

| How do **[participants/cases]** experience, interpret, or respond to **[phenomenon]** in **[context]**?

For a descriptive question:

| What is the distribution, pattern, or change in **[phenomenon]** among **[units]** in **[context/period]**?

Association is not causation

Associational question

- | How is AI use associated with students' grades?

Causal question

- | What is the effect of access to an AI tutor on later unaided test performance?

The causal wording implies a counterfactual comparison and a much stronger design requirement.

Avoid **affects, impacts, leads to, improves, reduces, or causes** unless the design can defend a causal interpretation.

Use FINER to test the question

Criterion	Ask yourself
Feasible	Can I complete it with my time, skills, access, and available evidence?
Interesting	Will the answer sustain my effort and interest a relevant audience?
Novel	What does it add, test, extend, or clarify?
Ethical	Can it be investigated without inappropriate risk or data use?
Relevant	Why does the answer matter theoretically, empirically, or practically?

See [Covvey et al. \(2024\)](#) for a discussion of FINER criteria.

Warning signs

- It is only a topic.
- It asks several major questions at once.
- It contains undefined judgments: “good,” “bad,” “effective,” “successful.”
- It presumes the answer.
- It uses causal language without a defensible comparison.
- It requires inaccessible, confidential, or nonexistent evidence.
- It is too broad for one semester.
- Its answer would add nothing beyond what is already known.

Evidence follows the question

Question	Plausible evidence
How do students experience AI-supported learning?	interviews, focus groups, or diaries
How common is weekly AI use?	a defensibly sampled survey or administrative records
How do university AI policies differ?	policy documents and systematic coding
Is AI use associated with grades?	measures of use, outcomes, and relevant covariates
Does access affect unaided performance?	experimental or credible quasi-experimental evidence

Convenient data are not necessarily appropriate evidence.

See the Appendix for primary versus secondary evidence and a detailed evidence-selection checklist.

Scope for this week

For your preliminary question, identify only:

- what evidence would be required in principle;
- whether a plausible source or collection route exists;
- the largest access, ethical, or measurement obstacle.

You do **not** need to download, scrape, clean, or analyze the final data this week.

| Literature searching begins in Week 3. Detailed data sourcing returns in Week 9.

Topic and problem

Topic

| Nonperforming loans in Ukrainian banks

Broad problem

| Ukrainian banks accumulated exceptionally high NPLs, threatening bank soundness and financial stability.

Research problem narrowed by the literature

| The role of related lending in the accumulation of bank-level NPLs—and the dynamic relationship between the two—required further investigation in the Ukrainian context.

The Appendix traces the questions, objectives, concepts, evidence, contribution, and bounded finding in detail.

The complete chain

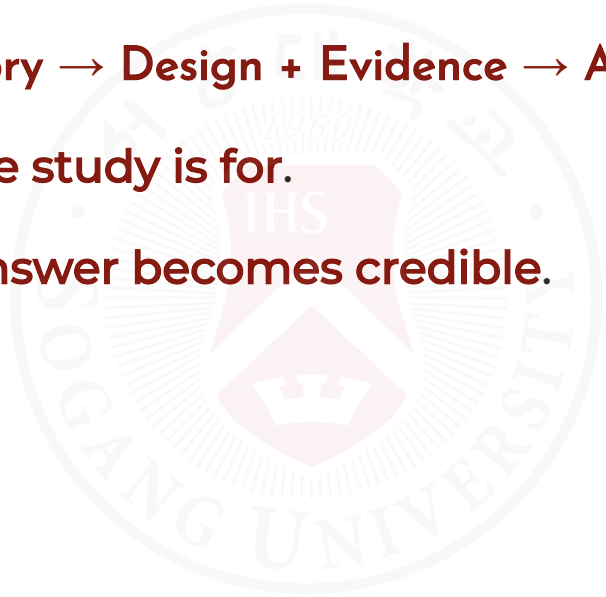
Topic → Problem → Question → Objective → Contribution

Then:

Concepts + Theory → Design + Evidence → Analysis → Claim

The first chain explains **what the study is for**.

The second explains **how the answer becomes credible**.



What to remember

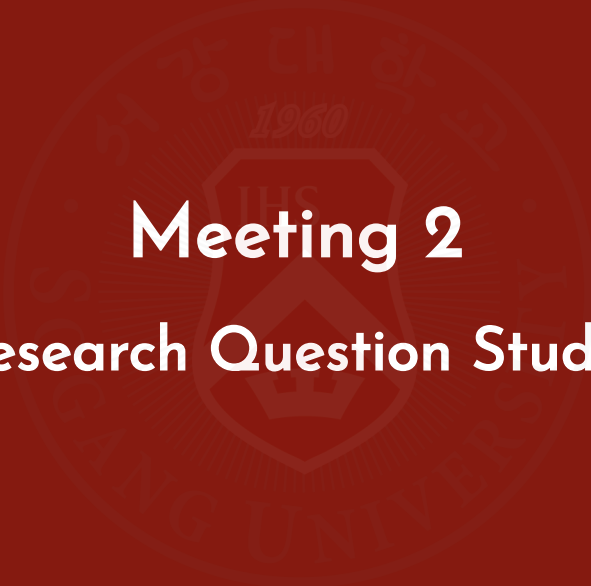
1. A topic is not a research question.
2. The problem explains why the question is needed.
3. Literature and theory help define the problem and expected contribution.
4. The question states what the study will answer.
5. The objective states what the study will do.
6. The contribution states what new value the answer could add.
7. Concepts, evidence, method, and claim must align with the question.
8. A feasible, modest question is better than an impressive but impossible one.

Prepare for Meeting 2

Bring:

- your Week 1 question seed;
- a laptop;
- one broad topic you genuinely care about;
- one possible problem or puzzle within it;
- any preliminary source, dataset, case, or context that inspired the idea.

You do not need a polished question. Meeting 2 is designed to create one.



Meeting 2

Research Question Studio

Agenda

Week 2 — Meeting 2

1. Rapid Recap
2. Two Pathways to a Research Topic
3. Individual Brainstorming and Memo
4. Small-Group Research Clinic
5. Individual Revision
6. Class Sharing
7. Assignment Submission and Next Week



Today's outcome

By the end of class, each student will submit:

- a working topic;
- a problem or puzzle;
- one preliminary research question;
- an aligned objective;
- a provisional contribution;
- key concepts requiring clarification;
- plausible evidence;
- a FINER self-assessment;
- one unresolved risk or uncertainty.

This is an **individual research project**. Peer work improves your reasoning, but the submitted decisions must be your own.

Reconstruct the chain – 3 minutes

Without looking at your notes, arrange these elements:

- contribution;
- research question;
- topic;
- objective;
- problem or puzzle.

Then explain what each transition accomplishes.



Pathway 1: from personal interest

1. **Topic:** I am studying _____
2. **Question:** because I want to find out what/why/how _____,
3. **Significance:** in order to help my reader understand _____.

Subject: central banking

Topic—I am studying: the effect of central-bank communication on financial markets

Question—because I want to find out: How does the clarity of central-bank communication affect market volatility?

Significance—in order to help my reader understand: how central-bank communication can influence market behavior and inform investment decisions

This is a promising starting formulation. If the eventual design is observational, revise **affect** to **is associated with** or justify a credible causal design.

See the Appendix for the Central Bankers Club origin story and intermediate focusing steps.

Pathway 2: from an interesting event

1. **Topic:** I am studying _____
2. **Question:** because I want to find out what/why/how _____,
3. **Significance:** in order to help my reader understand _____.

Starting observation: the global financial crisis

Topic—I am studying: the effectiveness of the Twin Peaks supervision model

Question—because I want to find out: How does the Twin Peaks model compare with other regulatory models in preventing financial crises?

Significance—in order to help my reader understand: how banking supervision and regulation might be improved to reduce future financial instability

This initial question is broad. The next step would be to define the comparison, cases, period, outcome, and what “preventing” means.

See the Appendix for the event images and intermediate focusing steps.

What the two pathways teach us

Research may begin with:

- a sustained personal or professional interest;
- an event that is surprising or consequential;
- a disagreement with existing explanations;
- a practical problem that exposes a conceptual problem.

But the starting inspiration is not enough. It must still become:

a focused topic → a research problem → a question → an objective → a contribution

Brainstorm and select – 7 minutes

Write freely. Do not edit yet.

1. What broad issue repeatedly interests or bothers you?
2. What have you observed that seems surprising, inconsistent, or insufficiently explained?
3. Who or what is affected?
4. In which place, institution, case, or period?
5. What would you genuinely like to know—not merely prove?
6. What evidence might exist?
7. Why might the answer matter?

Brainstorm and select – 7 minutes (continued)

Generate **three possible topic–problem–question directions**, then score them quickly:

Candidate	Interest	Feasibility	Possible contribution
1	1–3	1–3	1–3
2	1–3	1–3	1–3
3	1–3	1–3	1–3

Select one direction to develop today.

Do not automatically choose the most ambitious question. Choose the one you can turn into credible research.

Write your individual memo – 8 minutes

Write **1–2 paragraphs** explaining:

1. the subject and focused topic;
2. the practical or conceptual problem;
3. the condition and consequence/cost;
4. the research question;
5. why answering it may matter to readers;
6. the evidence you may need.

Then express the core in three phrases:

I am studying **[focused topic]**
because I want to find out **[research question]**
in order to help my reader understand **[significance]**.

Underline any statement that currently depends on an assumption rather than verified literature.

Complete the research chain – 4 minutes

Element	Your current version
Subject	broad area of knowledge
Topic	focused research territory
Problem	condition + consequence/cost
Question	precise answer sought
Objective	what the study will do
Contribution/significance	what the answer may add and why readers should care

Then add:

- **Purpose:** exploratory, descriptive, explanatory, causal, or predictive;
- **Approach:** likely qualitative, quantitative, or mixed methods;
- **Key concepts:** two or three terms requiring definition;
- **Evidence:** what would be needed in principle;
- **Claim boundary:** what the design probably could not establish.

These are provisional choices. The literature may change them.

Groups of 2-3 – 15 minutes

Share each memo and help rewrite the three core phrases.

For each student, ask:

1. Is the topic productively focused and expressed with an action word?
2. Is the problem a genuine gap in understanding—not merely a broad social concern?
3. Are both parts visible: **condition + consequence/cost**?
4. Does the question follow from the problem?
5. Does it invite a meaningful answer rather than merely yes/no?
6. Does it direct the researcher toward identifiable evidence?
7. Does the significance answer **So what?** for readers?
8. Is the project feasible in one semester?

Each student records the group's two most important suggested changes.

Feedback rule

Do not say only:

- “It is interesting.”
- “It is too broad.”
- “I like it.”

Give actionable feedback:

“Your question includes two outcomes. I suggest choosing **X** because your proposed evidence does not measure **Y**.”

“The question says **affect**, but the proposed cross-sectional data appear able to show only association.”

Instructor checkpoint during group work

Before final revision, be prepared to answer:

- What is the problem—not merely the topic?
- What exactly is the question?
- What will you do to answer it?
- What might the answer add?
- Which claim is beyond your likely design?
- What is your greatest feasibility risk?



FINER stress test – 3 minutes

Criterion	Score: 1–3	One-sentence justification
Feasible		
Interesting		
Novel		
Ethical		
Relevant		

Circle the lowest score and revise accordingly.

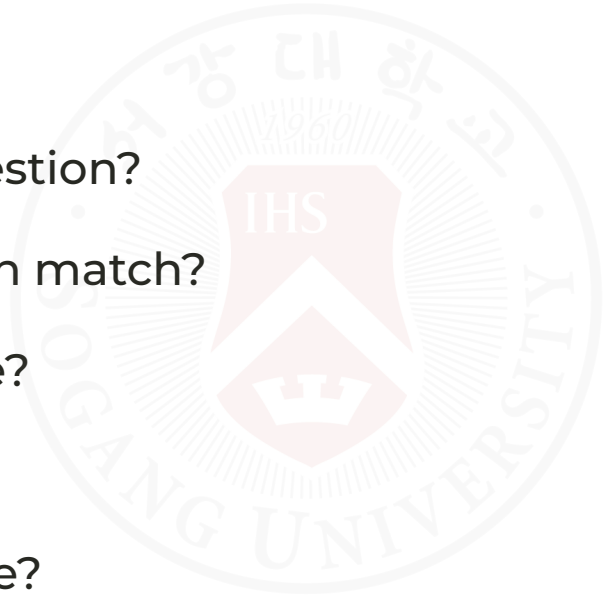
Do not claim novelty as a fact before the literature review. Write what you **expect to investigate or add**.

Final individual revision – 6 minutes

Revise your complete page using the feedback.

Final check:

- Is the problem clear?
- Is there exactly one main question?
- Do the objective and question match?
- Are the main concepts visible?
- Is causal language justified?
- Is plausible evidence available?
- Is the proposed contribution appropriately provisional?
- Can this be completed in one semester?



Share your three phrases – 8 minutes

Each student has approximately **30–40 seconds**:

I am studying ...
because I want to find out ...
in order to help my reader understand

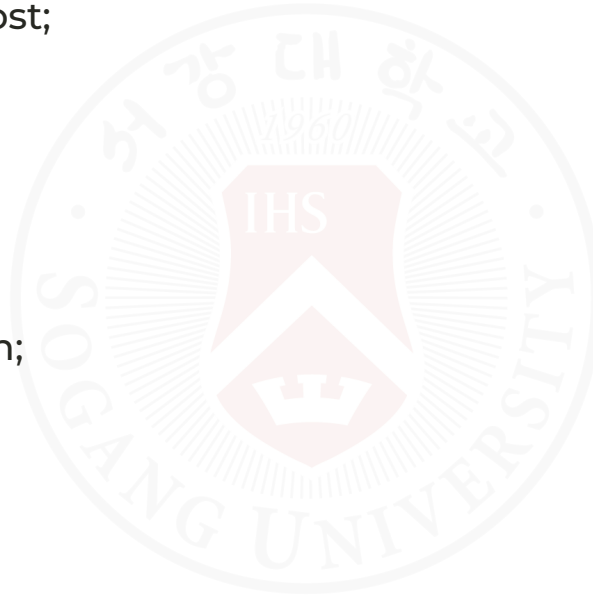
As you listen, identify:

- one especially clear question;
- one convincing statement of significance;
- one question that may need further narrowing.

Preliminary Research Question

Upload your individual memo to **Cyber Campus**. It must contain:

1. subject and focused topic;
2. problem: condition + consequence/cost;
3. preliminary research question;
4. research objective;
5. provisional contribution/significance;
6. intended purpose and likely approach;
7. key concepts requiring clarification;
8. plausible evidence and access route;
9. FINER self-assessment;
10. most important unresolved concern;
11. the final three-phase version.



What happens next?

This question is **preliminary**, not permanent.

In Week 3, you will use it to:

- construct literature-search terms;
- locate relevant scholarly work;
- assess what is already known;
- test whether the proposed contribution is credible;
- revise the problem, question, concepts, or scope.

Good research questions are developed through informed revision—not produced perfectly in one sitting.

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Next Week

Finding and evaluating literature

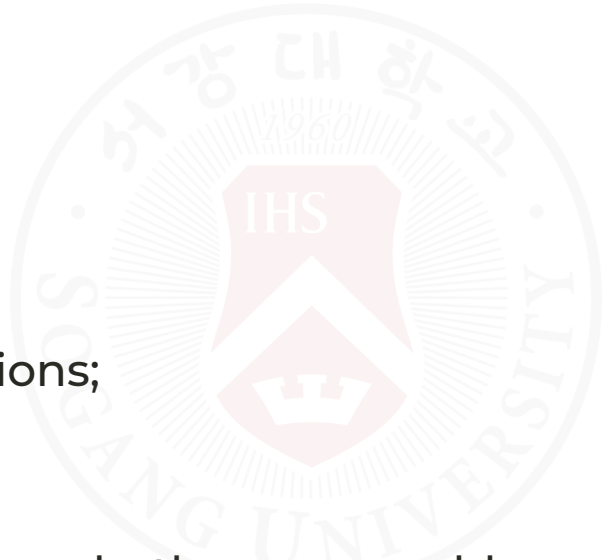
Prepare for Week 3

Bring your submitted preliminary question.

We will convert it into:

- core search concepts;
- synonyms and related terms;
- documented search strings;
- inclusion and exclusion decisions;
- a literature evidence matrix.

The literature will help determine whether your problem and proposed contribution are defensible.



Reading assignment: literature reviews

Read before Week 3:

1. Randolph, J. J. (2009). "A Guide to Writing the Dissertation Literature Review." *Practical Assessment, Research, and Evaluation*, 14, Article 13.
2. Obenzinger, H. (2005). *What Can a Literature Review Do for Me? How to Research, Write, and Survive a Literature Review*. Stanford University.

While reading, note:

- what a literature review should accomplish;
- how it differs from a list of summaries;
- one recommendation you could apply to your preliminary question.

The background of the slide features a large, faint, circular seal of Sogang University. The seal contains the university's name in Korean and English, the year 1960, and a central shield emblem.

Questions?

Thank you!



Appendix

Reference material

The background of the slide features a large, faint, circular seal of Yonsei University. The seal contains the university's name in Korean and English, the year 1960, and the letters IHS.

Appendix A

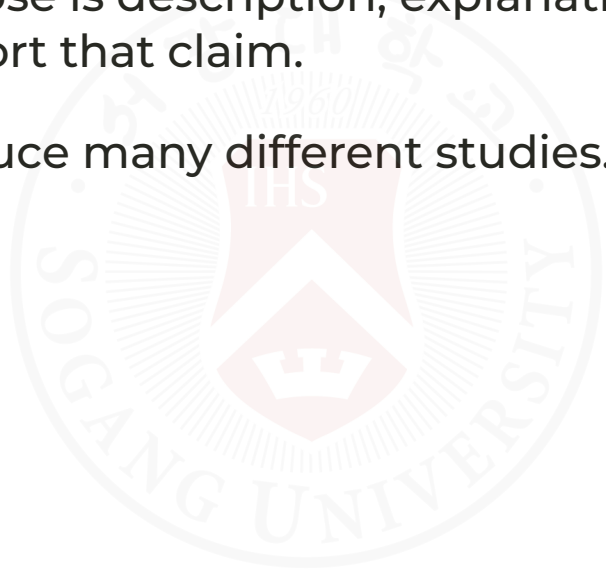
Extended diagrams and examples for Meeting 1

Return to the Week 1 claim

| “Students who use AI receive higher grades.”

To investigate it, define AI use, students, courses, period, and academic performance. Then decide whether the purpose is description, explanation, causation, or prediction—and what evidence could support that claim.

One broad statement can produce many different studies.



A practical research cycle

1. Identify an important problem or puzzle.
2. Examine relevant literature and theory.
3. Formulate and refine the research question.
4. Clarify concepts, objectives, and expected contribution.
5. Select a defensible design and appropriate evidence.
6. Obtain, document, and manage evidence ethically.
7. Analyze and challenge the interpretation.
8. Communicate claims, uncertainty, and limitations.
9. Revise and preserve a reproducible record.

Question-to-objective example

Question

How is daily social-media use associated with self-reported anxiety among Sogang undergraduates?

Objective

To estimate and interpret the association between daily social-media use and self-reported anxiety in a sample of Sogang undergraduates.

Use verbs that match the purpose: **describe, explore, compare, explain, estimate, evaluate, or predict.**

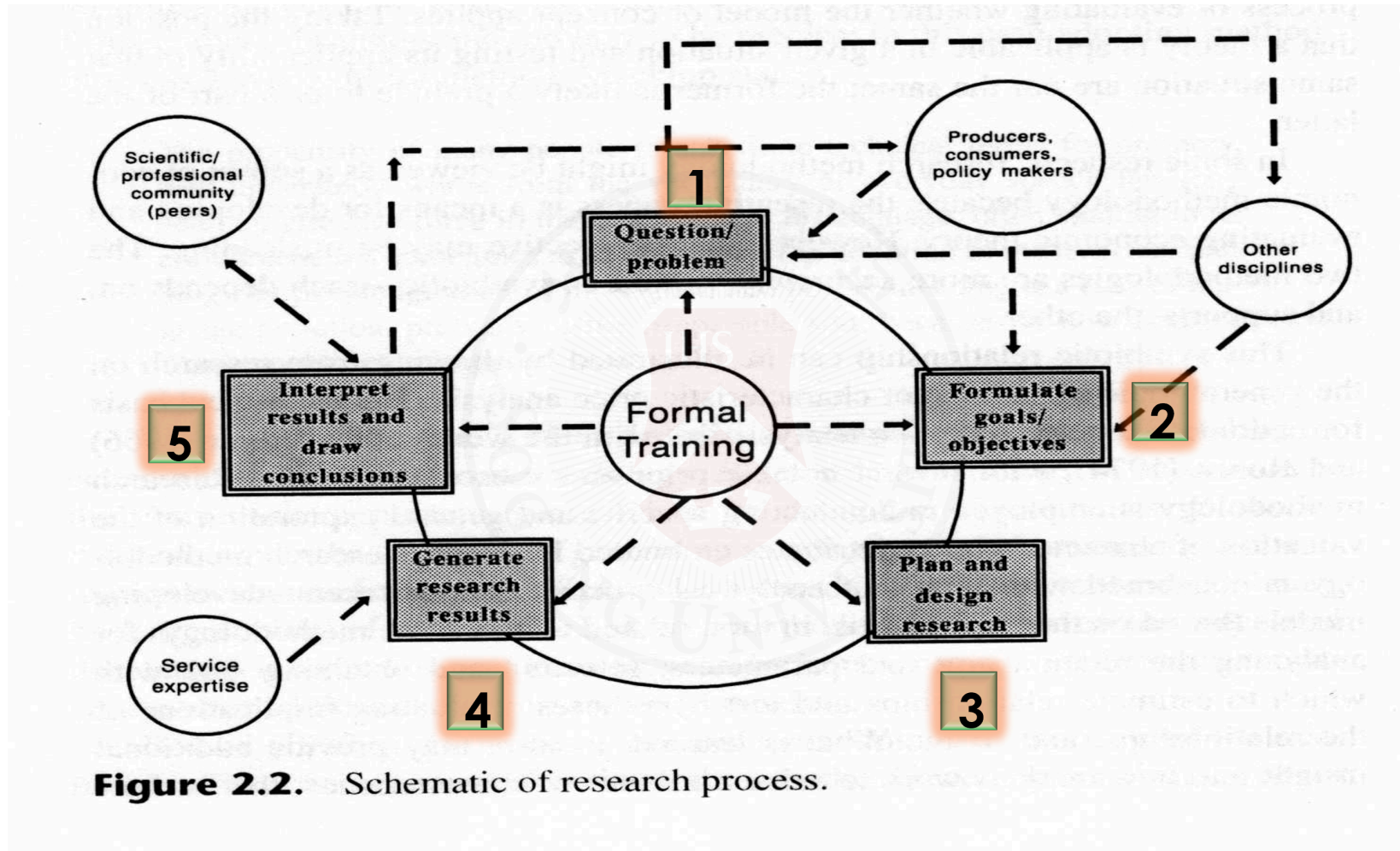
Types of research contribution

A study may contribute:

- evidence from an understudied context;
- a better measure, dataset, or design;
- evidence about a mechanism or subgroup;
- a test of whether an established finding travels;
- integration of previously separated perspectives;
- a carefully documented contradictory or null result.

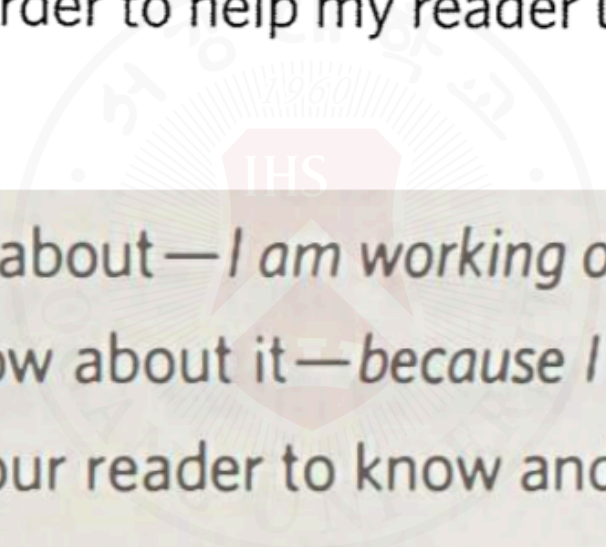
Contribution is always judged **relative to existing literature**.

Alternative research-process diagram



Topic, question, and significance

1. **Topic:** I am studying _____
2. **Question:** because I want to find out what/why/how _____,
3. **Significance:** in order to help my reader understand _____.

- 
1. what you are writing about — *I am working on the topic of...*
 2. what you don't know about it — *because I want to find out...*
 3. why you want your reader to know and care about it — *in order to help my reader understand better...*

Conceptual problem: visual structure

If we can't answer the question of how romantic movies have changed in the last fifty years, *condition / first question* **then we can't answer a more important question: How have our cultural depictions of romantic love changed?** *consequence / larger, more important second question*

If we can't answer the question of how our depictions of romantic love have changed, *second question* **then we can't answer an even more important one: How does our culture shape the expectations of young men and women about marriage and families?** *consequence / larger, more important question*

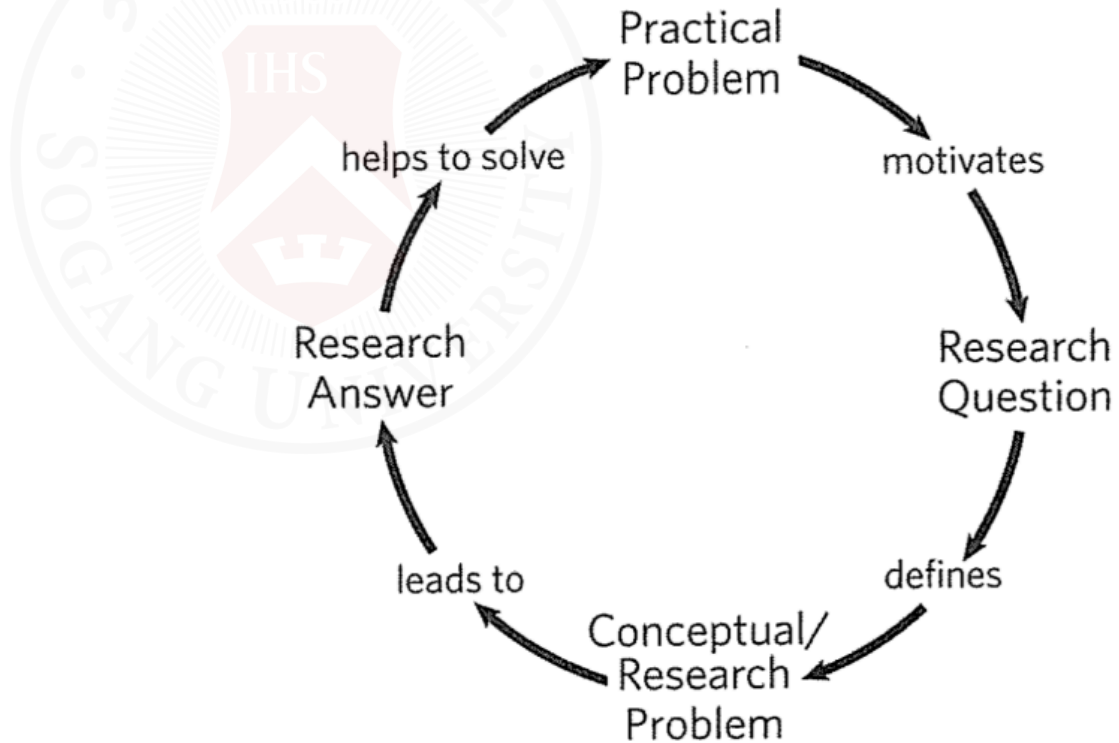
Conceptual and practical problems

PRACTICAL PROBLEM: The chain on my bicycle broke.

RESEARCH PROBLEM: Can I find a bike shop that will replace it?

RESEARCH SOLUTION: Here it is: Cycle Source, 1401 East 55th Street.

PRACTICAL SOLUTION: Walk over to get my bike fixed.



Why problem definition matters

1. **Topic:** I am studying _____
2. **Question:** because I want to find out what/why/how _____,
3. **Significance:** in order to help my reader understand _____.

Without a clear problem and significance, research becomes data collection without a contribution.

Primary and secondary evidence

Type	Meaning	Examples
Primary	collected by you or your team for the present study	your survey, interview, observation, experiment
Secondary	previously collected by someone else or for another purpose, then reused	official statistics, archives, existing surveys, administrative records

The distinction depends on your relationship to the original collection—not the file format.

Evidence-selection checklist

Ask whether the evidence:

1. represents the concepts, population, context, and period;
2. has clear provenance and production methods;
3. has important omissions or exclusions;
4. can be used under applicable permissions, licenses, and ethical rules;
5. supports the intended claim;
6. permits a documented and reproducible workflow.

Adapted from the [UK Data Service](#).

Worked example: questions and objectives

Research questions represented by the study

1. How are related lending and other bank-specific factors associated with banks' NPLs in Ukraine?
2. How do shocks to NPLs relate dynamically to subsequent related lending?

Corresponding objectives

1. Estimate the relationship between related lending, bank-specific factors, and NPLs.
2. Examine the dynamic response of related lending to shocks in NPLs.

Worked example: concepts, theory, and evidence

Concepts: nonperforming loans, related lending, bank soundness, risk-taking.

Theoretical tension: relationships may provide an information advantage, while conflicts of interest may weaken screening and monitoring.

Evidence and design:

- unbalanced quarterly panel of 207 Ukrainian banks;
- Q2 2008–Q4 2020;
- bank-level quantitative evidence and dynamic analysis.

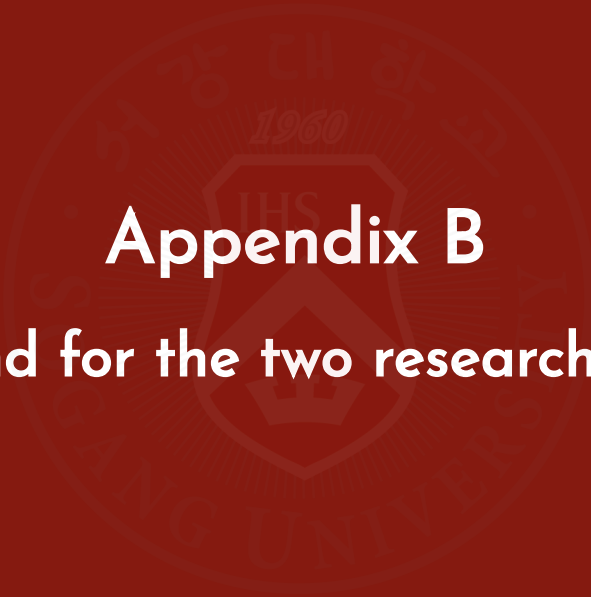
Worked example: contribution and bounded claim

Contribution: evidence from an understudied institutional context, explicit attention to related lending, and analysis of the relationship in both directions.

Bounded finding

NPLs increased with related lending, while a positive shock to NPLs substantially increased related lending.

Vyshnevskyi, I., & Sohn, W. (2023), “Nonperforming loans and related lending: Evidence from Ukraine,” *Emerging Markets Review*, 57, 101069. [Abstract and citation.](#)



Appendix B

Background for the two research pathways

Personal-interest origin: Central Bankers Club



From personal interest to research

Emerging Markets Review 59 (2024) 101121



ELSEVIER

Contents lists available at [ScienceDirect](https://www.sciencedirect.com)

Emerging Markets Review

journal homepage: www.elsevier.com/locate/emr



The clarity of monetary policy communication and financial market volatility in developing economies

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ABSTRACT

This study investigates the effect of the clarity of monetary policy statements (MPSs) on financial market volatilities by employing linguistic analysis and a panel fixed effects estimator for a newly built dataset drawn from the MPSs of 21 developing countries. Our results show that MPS clarity, measured by complexity and readability indices, negatively affects foreign exchange rate volatility, a key policy variable in developing economies, suggesting that clear central bank communication can reduce financial market volatility. Given the low literacy levels in developing countries, their central banks should make an extra effort to write statements that are easy to comprehend.

From interest to a focused direction

This study differs from previous research in three ways. First, most studies have been conducted in advanced and emerging market economies, in which financial markets are well or fairly well developed. Developing countries with less efficient financial markets are generally slow to react to news and, at times, do not respond at all. We investigate, therefore, whether the impact of major monetary policy communication on financial markets in advanced economies can be observed in the developing world.

Second, we analyze monetary policy statements (MPSS) officially released by central banks, which are among the most important monetary policy communication tools. Because most central banks in the developing world began issuing monetary policy statements (MPSS) in the last decade, it is now possible to investigate the impact of official monetary policy communication on financial markets.

Third, this study constructs a novel dataset comprising more than 850 statements from 21 central banks in developing countries. Using these statements, we measure both complexity and readability indices widely used in the linguistics literature and employ a panel fixed effects estimator to explore the influence of central bank communication on financial market volatility in developing countries.

The central question of this study is whether the clarity of central bank communication affects financial market volatility in developing countries. Our hypothesis is that the longer and less clear (measured through complexity and readability) the statements are, the larger the volatility in financial markets as measured by foreign exchange rates, lending rates, money market rates, and treasury bill rates. Longer sentences in policy statements may have a higher likelihood of relaying unclear information and present a higher probability of market players having a large variance in their judgments regarding current and prospective policies. We also expect unclear statements to have an unfavorable effect on financial markets because MPS uncertainty may cause additional financial market volatility.

We find a negative and statistically significant relationship between MPS clarity and foreign exchange (FX) rate volatility. These results suggest that central bank communication in developing countries influences financial market volatility to a certain extent, as it does in advanced and emerging market economies, and that easy and clear central bank communication can reduce financial market volatility. Therefore, simple MPSS can be used as an effective monetary policy communication tool. Given the literacy levels in developing nations, where their monetary policy may be poorly comprehended, complex and unclear MPSS can cause unnecessary and considerable volatilities in the financial markets, suggesting that their central banks should put extra effort into making their statements easy to understand.

This study contributes to the literature by constructing a newly created dataset drawn from MPSS to quantify their complexity and readability and focuses on the monetary policy communication of developing countries.

Event origin: the global financial crisis



From event to research direction

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The 'Twin Peaks' model of post-crisis banking supervision

Wook Sohn ✉ & Iegor Vyshnevskiy

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ABSTRACT

This article examines the effectiveness of the 'Twin Peaks' supervision model, in which consumer protection and prudential regulation are carried out by two separate regulatory agencies. We use a dataset from 143 countries for the period of 2004–2011 to find that the Twin Peaks model is associated with a higher quality of banking supervision in terms of the efficiency and stability of financial institutions.

Q KEYWORDS: Banking supervision framework The 'Twin Peaks' model financial regulatory reform global financial crisis

QJEL CLASSIFICATION: G21 G28

From event to research direction

I. Introduction

The regulatory framework for financial stability has been much debated along with the need to tighten regulation of banking activities and balance sheets following the global financial crisis. Financial regulatory agencies have appeared at the forefront of these regulatory reforms, partly due to their failures during the crisis and partly because, in those days, their regulatory structure did not help to endure the crisis.

Working Group on Financial Reform of Group of Thirty (2008) report observed that although countries were dealing with similar financial issues, their regulatory systems and approaches varied. The report classifies financial supervision into the sectoral, functional, integrated and 'Twin Peaks' frameworks. The Twin Peaks regulatory framework is considered to be one in which there is a separation of the safety and soundness objective and the conduct-of-business regulation between two regulatory agencies. The report argues that 'The Twin Peaks approach may be the optimal means of ensuring that issues of transparency, market integrity, and

Development (OECD, 2009) also supports the above assertion. Moreover, the European Central Bank (2010) has identified member countries – namely the Netherlands, Italy, Belgium, the United Kingdom, Portugal and France – that have either already moved to the full Twin Peaks model or intend to use some elements of it.

The Twin Peaks model is considered as one of the most effective ways of managing financial systems, as countries implementing this regulatory model, such as Australia and the Netherlands, were the least hit by the global financial crisis. However, analytical evidence concerning the model's performance will answer the question of whether this system is the best response to the current challenges in the financial system.² The objective of this article is to examine the empirical evidence on this topic.

II. Empirical analysis

We use data on banking supervision quality, as well as on the structure of banking supervision, for 143 countries in the period of 2004–2011. All the relevant data are taken from the World Bank database

Research classifications: keep dimensions separate

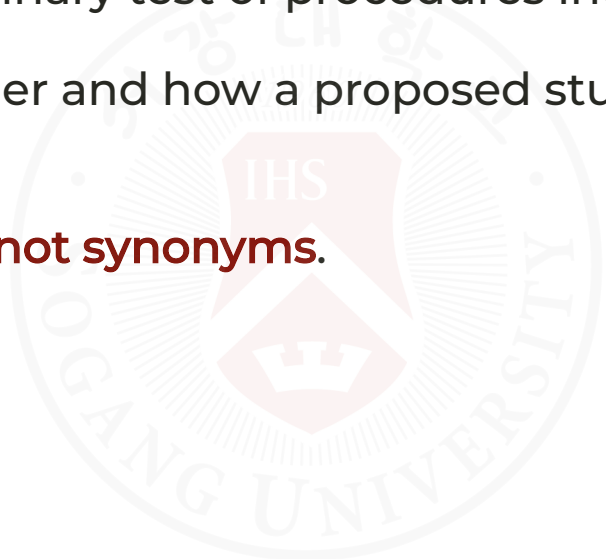
Dimension	Examples
Purpose	exploratory, descriptive, explanatory, causal, predictive
Approach	qualitative, quantitative, mixed methods
Knowledge aim	theoretical, empirical; basic, applied
Time structure	cross-sectional, time series, repeated cross-section, panel/longitudinal
Design/method	experiment, survey, case study, interview, observation, document analysis
Evidence origin	primary, secondary

These labels may be combined. For example: an **applied, quantitative, explanatory study using secondary panel data**.

Exploratory research and pilot studies

- **Exploratory research** investigates a poorly understood phenomenon and helps clarify concepts, questions, mechanisms, or hypotheses.
- A **pilot study** is a small preliminary test of procedures intended for a later, larger study.
- A **feasibility study** asks whether and how a proposed study or intervention can be carried out.

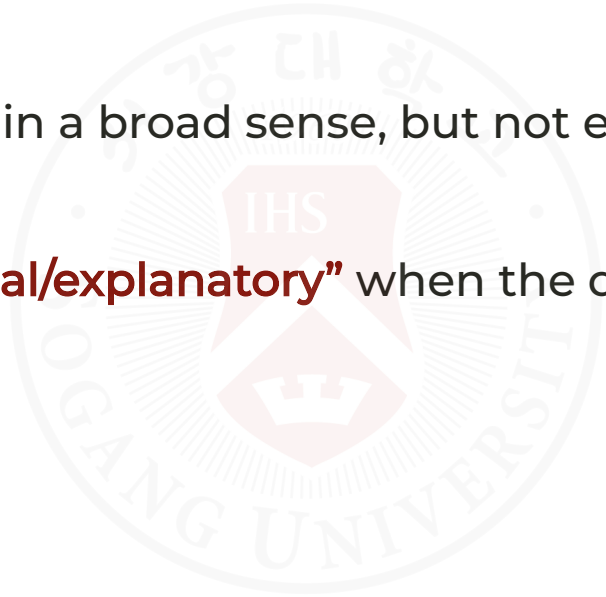
They may overlap, but they are **not synonyms**.



Explanatory and causal questions

- **Explanatory research** seeks to understand why or how a phenomenon occurs.
- **Causal research** seeks to estimate what would change under a different treatment, exposure, or intervention.
- A causal study is explanatory in a broad sense, but not every explanatory study identifies a causal effect.

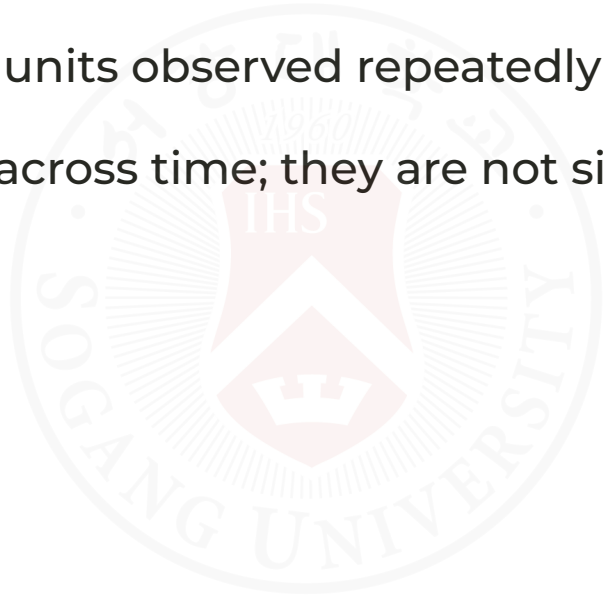
Avoid the combined label **“causal/explanatory”** when the distinction matters.



Time structures

- **Cross-sectional:** multiple units observed at one time or within a short reference period.
- **Time series:** repeated observations on one aggregate unit or variable over time.
- **Panel/longitudinal:** the same units observed repeatedly over time.

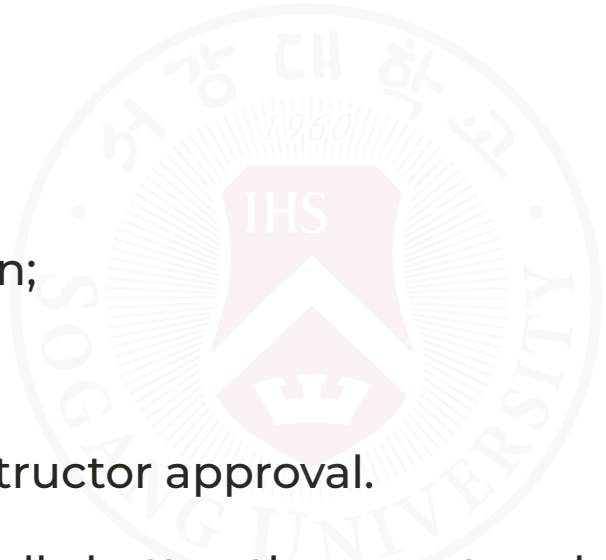
Panel data track multiple units across time; they are not simply another name for time-series data.



Primary evidence: main considerations

- tailored measurement;
- sampling and recruitment;
- informed consent;
- participant burden and risk;
- privacy and data minimization;
- secure storage and access;
- required ethical review or instructor approval.

Primary data are not automatically better; they create substantial design and ethical responsibilities.



Secondary evidence: main considerations

- provenance and original purpose;
- population and temporal coverage;
- measurement definitions and changes;
- missingness, revision, and selection;
- license and access conditions;
- privacy and disclosure risk;
- fit with the new research question.

Even public data require critical evaluation and proper citation.

Source notes

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